Evaluation runs: **memcached** (memcached-oopsla25)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2	3	3	violation	0.0	2	1 h, 250 GB	3	9	brain03, 2026-11-02
3	7	7	violation	0.0	2	1 h, 250 GB	7	49	brain03, 2026-11-02

stateless: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1	1	2	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
2	3	6	conforms	0.0	2	1 h, 250 GB	3	6	brain03, 2026-11-02
3	7	20	conforms	0.0	2	1 h, 250 GB	7	28	brain03, 2026-11-02

stateless: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
2	3	3	conforms	0.0	2	1 h, 250 GB	3	6	brain03, 2026-11-02
3	7	7	conforms	0.0	2	1 h, 250 GB	7	28	brain03, 2026-11-02

Not run (stateless). the tool ended with an error (exit-101): 4 lost-set fifo, 4 shared-bucket fifo, 4 v3 fifo.

skipped: a smaller size had reached its limit: 5 shared-bucket fifo, 5 lost-set fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 shared-bucket fifo, 7 lost-set fifo, 7 v3 fifo, 8 shared-bucket fifo, 8 lost-set fifo, 8 v3 fifo, 9 shared-bucket fifo, 9 lost-set fifo, 9 v3 fifo.

stateful

stateful: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
2	3	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	7	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	1	2	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
2	3	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	7	20	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
2	3	3	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	7	7	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). the tool ended with an error (exit-101): 4 lost-set fifo, 4 shared-bucket fifo, 4 v3 fifo.

skipped: a smaller size had reached its limit: 5 shared-bucket fifo, 5 lost-set fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 shared-bucket fifo, 7 lost-set fifo, 7 v3 fifo, 8 shared-bucket fifo, 8 lost-set fifo, 8 v3 fifo, 9 shared-bucket fifo, 9 lost-set fifo, 9 v3 fifo.

offline

offline: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	3	3	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	7	7	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02

offline: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	2	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2	3	6	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	7	20	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
2	3	3	conforms	0.3	8	10 min, 16 GB	0.0	0.3	0.0	in-memory	brain05, 2026-11-02
3	7	7	conforms	0.2	8	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain05, 2026-11-02

Not run (offline). skipped: a smaller size had reached its limit: 4 lost-set fifo, 4 shared-bucket fifo, 4 v3 fifo, 5 shared-bucket fifo, 5 lost-set fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 shared-bucket fifo, 7 lost-set fifo, 7 v3 fifo, 8 shared-bucket fifo, 8 lost-set fifo, 8 v3 fifo, 9 shared-bucket fifo, 9 lost-set fifo, 9 v3 fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	175	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-02
2	3	3	violation	3.1	176	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	7	7	violation	3.4	174	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	2	conforms	2.7	162	10 min, 16 GB	0.1	2.7	0.0	in-memory	brain01, 2026-11-02
2	3	6	conforms	3.8	166	10 min, 16 GB	0.0	3.8	0.0	in-memory	brain01, 2026-11-02
3	7	20	conforms	4.3	162	10 min, 16 GB	0.0	4.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	156	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-02
2	3	3	conforms	3.1	170	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	4.3	178	10 min, 16 GB	0.0	4.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	156	10 min, 16 GB	0.6	0.1	0.0	in-memory	brain01, 2026-11-02
2	3	3	violation	2.9	158	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain01, 2026-11-02
3	7	7	violation	3.1	176	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	2	conforms	0.6	156	10 min, 16 GB	0.6	0.0	0.0	in-memory	brain01, 2026-11-02
2	3	6	conforms	2.9	158	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain01, 2026-11-02
3	7	20	conforms	3.1	176	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	156	10 min, 16 GB	0.6	0.3	0.0	in-memory	brain01, 2026-11-02
2	3	3	conforms	2.9	158	10 min, 16 GB	2.9	0.3	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	3.1	176	10 min, 16 GB	3.1	0.2	0.0	in-memory	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	0.6	152	1 h, 250 GB		brain01, 2026-11-02
2	3	3	violation	2.9	154	1 h, 250 GB	1	brain01, 2026-11-02
3	7	7	violation	3.1	172	1 h, 250 GB	2	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair shared-bucket, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	2	conforms	0.6	152	1 h, 250 GB		brain01, 2026-11-02
2	3	6	conforms	2.9	154	1 h, 250 GB		brain01, 2026-11-02
3	7	20	conforms	3.1	172	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair v3, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	0.6	152	1 h, 250 GB		brain01, 2026-11-02
2	3	3	conforms	2.9	154	1 h, 250 GB		brain01, 2026-11-02
3	7	7	conforms	3.1	172	1 h, 250 GB		brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair lost-set, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	175	10 min, 16 GB	0.6	0.1	0.0	in-memory	brain01, 2026-11-02
2	3	3	conforms	3.2	176	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	3.4	173	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair shared-bucket, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.7	162	10 min, 16 GB	2.7	0.0	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.9	165	10 min, 16 GB	3.8	0.0	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	4.3	162	10 min, 16 GB	4.3	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair v3, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.7	156	10 min, 16 GB	0.6	0.3	0.1	in-memory	brain01, 2026-11-02
2	3	3	conforms	3.2	170	10 min, 16 GB	3.1	0.3	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	4.3	178	10 min, 16 GB	4.3	0.2	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair lost-set, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	175	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-02
2	3	3	conforms	3.1	176	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	3.4	173	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.7	162	10 min, 16 GB	0.0	2.7	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.8	165	10 min, 16 GB	0.0	3.8	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	4.3	162	10 min, 16 GB	0.1	4.3	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	156	10 min, 16 GB	0.3	0.6	0.0	in-memory	brain01, 2026-11-02
2	3	3	conforms	3.1	170	10 min, 16 GB	0.3	3.1	0.0	in-memory	brain01, 2026-11-02
3	7	7	conforms	4.3	178	10 min, 16 GB	0.2	4.3	0.0	in-memory	brain01, 2026-11-02

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	160	10 min, 16 GB	0.6	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.7	168	10 min, 16 GB	2.7	0.2	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.7	152	10 min, 16 GB	0.6	0.2	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 2 fifo shared-bucket, 2 fifo lost-set, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.6	160	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.7	168	10 min, 16 GB	0.2	2.7	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	1	conforms	0.7	152	10 min, 16 GB	0.2	0.6	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 2 fifo shared-bucket, 2 fifo lost-set, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	0.8	149	1 h, 250 GB		brain02, 2026-11-03
2	3		failed			1 h, 250 GB		brain02, 2026-11-03
3	7		failed			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **shared-bucket**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.5	152	1 h, 250 GB		brain02, 2026-11-03
2	6		failed			1 h, 250 GB		brain02, 2026-11-03
3	20		failed			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **v3**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	0.8	152	1 h, 250 GB		brain02, 2026-11-03
2	3		failed			1 h, 250 GB		brain02, 2026-11-03
3	7		failed			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **lost-set**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	2.8	156	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair shared-bucket, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	2.7	164	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair v3, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	1	conforms	0.6	148	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). skipped: a smaller size had reached its limit: 2 fifo shared-bucket, 2 fifo lost-set, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.